Cisco PE Node MAR-PE-13 Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) Standard Operating Procedure (SOP)

Node: MAR-PE-13 | **Vendor:** Cisco | **Severity:** Major | **Alarm Code:** ALARM_4037

Description

The failure domain Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) in the Cisco PE node MAR-PE-13 at the Marathahalli Core site is primarily due to the degradation of the SFP/XFP transceivers or the occurrence of fiber cuts resulting in loss of optical signal power. This could be attributed to the inherent properties of optical fiber communication where signal attenuation increases with distance or due to physical damage to the optical fibers. The Cisco PE node, as a critical network component, plays a pivotal role in aggregating and transmitting high-speed data traffic across the optical transport network. Therefore, any failure in the optical transport domain can have a cascading effect on the overall network performance and availability. The alarm code ALARM_4037, which is associated with this failure domain, indicates a major issue that requires immediate attention from the network engineering team.

Pre-requisite Checks

- Pre-requisite: Verify that the site location, Marathahalli Core, is accessible and safely approachable for the network engineer, Status: Checked

- Pre-requisite: Ensure that the redundant link status is verified and functioning as expected, Status: Checked

- Pre-requisite: Confirm that the necessary tools and equipment are available for safe and efficient troubleshooting, Status: Checked

- Pre-requisite: Obtain the necessary permissions and clearances for access to the network equipment, Status: Checked

Potential Root Causes

Cause	Probability
SFP/XFP transceiver degradation	High
Fiber cuts or physical damage	Medium
Rx/Tx power loss due to optical signal attenuation	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface POS 0/3/0, Expected Output: Port 0/3/0 is up (line protocol is up), Description: This command verifies the status of the POS interface 0/3/0 and confirms whether it is up and
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functioning as expected.

Command: show interface POS 0/3/0 transceiver, Expected Output:
Transceiver is present and functioning, Description: This command
provides information about the transceiver present on the POS
interface 0/3/0 and confirms whether it is functioning as expected.

Command: show log | include ALARM_4037, Expected Output: ALARM_4037:
Major Optical Transport Failure, Description: This command searches
the system log for any occurrences of the alarm code ALARM_4037 and
provides information about the timestamp, severity, and description of
the alarm.
```

Resolution Steps

Step 1: Step: Inspect the SFP/XFP transceiver, Command: show interface POS 0/3/0 transceiver detail, Description: This step involves visually inspecting the SFP/XFP transceiver present on the POS interface 0/3/0 for any signs of physical damage, corrosion, or contamination. The command provided checks the detailed information about the transceiver and confirms whether it is functioning as expected.

Step 2: Step: Replace the SFP/XFP transceiver, Command: interface POS 0/3/0, Description: This step involves replacing the faulty SFP/XFP transceiver with a new one. The command provided configures the POS interface 0/3/0 and prepares it for the replacement of the transceiver.

Step 3: Step: Verify the optical signal power, Command: show interface POS 0/3/0 transceiver, Description: This step involves verifying the optical signal power of the POS interface 0/3/0 after replacing the SFP/XFP transceiver. The command provided checks the information about the transceiver and confirms whether it is functioning as expected.

Post-Check Verification

```
Command: show interface POS 0/3/0, Expected Output: Port 0/3/0 is up
(line protocol is up), Description: This command verifies the status
of the POS interface 0/3/0 and confirms whether it is up and
functioning as expected.

Command: show log | include ALARM_4037, Expected Output: No occurrence
of ALARM_4037, Description: This command searches the system log for
any occurrences of the alarm code ALARM_4037 and confirms that the
issue has been resolved.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP transceiver, Command: interface POS 0/3/0, Description: This step involves restoring the original SFP/XFP transceiver after replacing it with a new one. The command provided configures the POS interface 0/3/0 and prepares it for the restoration of the original transceiver.

- Step: Re-verify the optical signal power, Command: show interface POS 0/3/0 transceiver, Description: This step involves re-verifying the optical signal power of the POS interface 0/3/0 after restoring the original SFP/XFP transceiver. The command provided checks the information about the transceiver and confirms whether it is functioning as expected.